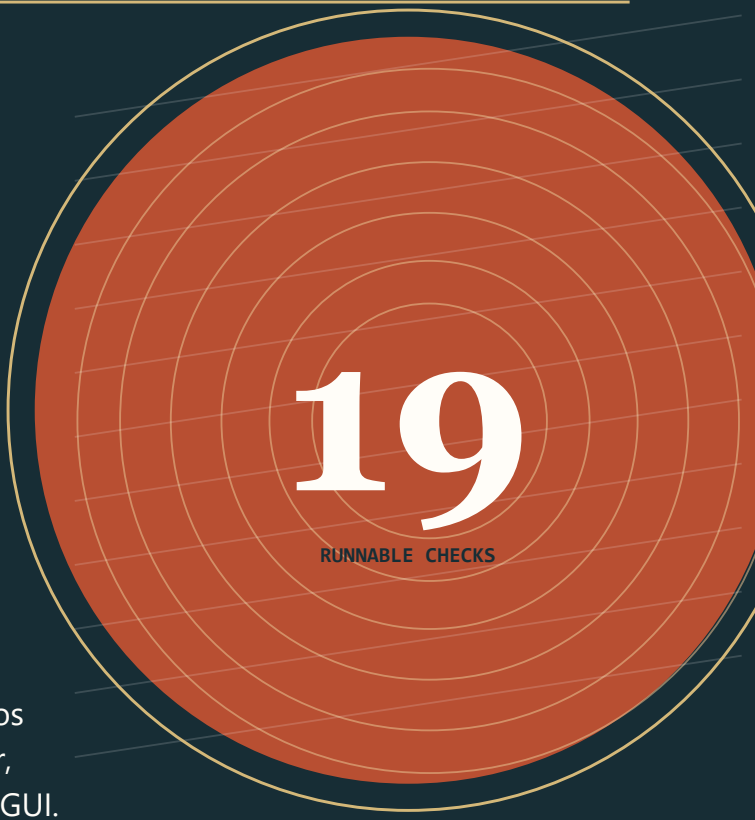
TEST PLAYBOOK / REV. 01

Mission control, tested.



A practical English-language guide to the scenarios already available in the MARS-OPS Testing Center, plus hands-on routes through the administration GUI.

IN THIS FIELD GUIDE

19 built-in test scenarios

5 real GUI walkthroughs

Expected outcomes and checks

START HERE / 4 STEPS

A test run, without the guesswork

The Scenario Lab runs each check against a fresh test colony and shows whether the observed result matches the expected result.

01

OPEN

Go to Testing center from the left navigation.

02

CHOOSE

Pick a category, then select a test. Review What you need and Expected result before running it.

03

RUN

Press Run this test. Use Run this category or Run all tests for a broader check.

04

CHECK

Read PASS or FAIL, then select the result row for the actual result and any technical details.

READ THE RESULT

PASS

The check reached its stated expected condition.

FAIL

Compare the Actual result with the expected result. Select the row to inspect technical details.

Each built-in scenario creates a fresh fixture. Running tests does not use the live colony objects.

AVAILABLE FILTERS

All tests	**Agent tests**	**Mission tests**	**Resource tests**
Facility tests	**Emergency tests**	**Validation tests**	**Edge case tests**

SESSION NOTE

Administration changes are kept in memory for the current GUI session and are not saved after closing the application.

01-05 / MISSION INTAKE

Mission intake / safe submission

Start with valid data, then see how the colony protects assignments from invalid teams and resources.

01

MISSION / VALID SUBMISSION

Valid exploration submission

TEST SETUP

A scientist, an equipped exploration rover, and enough oxygen are available. Select an exploration mission that requires 2 units of oxygen.

EXPECTED RESULT

The colony accepts it. The mission is ASSIGNED and becomes the active mission.

02

VALIDATION / AVAILABILITY

Unavailable agent is rejected

TEST SETUP

The selected scientist is already unavailable. Submit an exploration mission with that scientist.

EXPECTED RESULT

Submission is rejected. The mission is FAILED; the scientist stays unavailable.

03

VALIDATION / TEAM SKILLS

Unsuitable personnel is rejected

TEST SETUP

Select a mechanical engineer with no scientific skill for an exploration mission.

EXPECTED RESULT

Submission is rejected, the mission is FAILED, and no active mission is created.

04

VALIDATION / DUPLICATES

Duplicate agent is rejected

TEST SETUP

Add the same scientist twice to one mission team.

EXPECTED RESULT

Submission is rejected. The scientist remains available and the mission is FAILED.

05

MISSION / ROBOT ABILITY

Missing robot capability is rejected

TEST SETUP

Use a rover with surface travel and equipment carrying, but without EXPLORATION.

EXPECTED RESULT

The mission is rejected and marked FAILED because the required ability is missing.

06-10 / TEAM & CONDITIONS

Team integrity / operating conditions

These checks cover resource limits, the single active-mission rule, rescue team composition, and environment gating.

06

RESOURCES / SAFE REJECTION

Insufficient resources are not consumed

TEST SETUP

Set habitat oxygen to 2 units; submit a mission requiring 5 units.

EXPECTED RESULT

The mission is rejected. Habitat oxygen remains 2 units.

07

MISSION / SINGLE ACTIVE SLOT

One active mission at a time

TEST SETUP

Submit one valid exploration mission, then try to submit another before the first ends.

EXPECTED RESULT

The first mission remains active. The second submission is rejected and the second mission is FAILED.

08

RESCUE / INCOMPLETE TEAM

Rescue team composition is enforced

TEST SETUP

Submit a rescue mission with a capable surface vehicle but no communication-skilled personnel.

EXPECTED RESULT

The rescue is rejected as an incomplete team; its state is FAILED.

09

RESCUE / VALID TEAM

Valid Rescue team is accepted

TEST SETUP

Select a communication-skilled engineer and a vehicle with surface-travel capability.

EXPECTED RESULT

The colony accepts the rescue mission and marks it ASSIGNED.

10

ENVIRONMENT / EXPLORATION

Dust storm blocks exploration

TEST SETUP

Set the environment to DUST_STORM, then submit a valid exploration team.

EXPECTED RESULT

The mission is rejected with an environment-related failure reason and marked FAILED.

11-15 / MISSION LIFECYCLE

Lifecycle / time gates

Verify the allowed path from assignment through completion or failure, including invalid transitions.

LIFECYCLE / COMPLETION

11 Successful mission completes and releases agents

TEST SETUP

Submit and start a valid exploration mission, then advance simulation time by its 1-unit duration.

EXPECTED RESULT

The mission becomes COMPLETED. Its agents become available and the active-mission slot clears.

LIFECYCLE / FAILURE

12 Failed active mission releases agents

TEST SETUP

Submit and start a valid mission, then record a failure through Colony.

EXPECTED RESULT

The mission becomes FAILED. Its agents become available and the active-mission slot clears.

LIFECYCLE / START GUARD

13 Mission cannot start before association

TEST SETUP

Create a new mission and call startMission before the colony accepts it.

EXPECTED RESULT

The domain method throws IllegalStateException; the scenario records this as the expected result.

LIFECYCLE / COMPLETION GUARD

14 Mission cannot complete before it starts

TEST SETUP

Submit a mission, but call finish while it is still ASSIGNED.

EXPECTED RESULT

IllegalStateException is expected. The mission remains assigned and active.

EMERGENCY / RESPONSE

15 Oxygen leak uses real emergency response

TEST SETUP

Use a habitat with 10 oxygen and 5 spare parts; respond with a mechanical engineer and repair robot.

EXPECTED RESULT

The emergency resolves; oxygen becomes 9 and spare parts become 4. Responders are released.

16-19 / EMERGENCY & INPUTS

Emergency response / input guards

Check emergency facility validation, resource input validation, and simulation time limits.

16

FACILITY / TYPE CHECK

Wrong facility type is rejected

TEST SETUP

Try to create a Fire emergency against a Habitat instead of the required Laboratory.

EXPECTED RESULT

The Fire constructor rejects the facility with IllegalArgumentException.

17

RESOURCE / INPUT CHECK

Invalid resource amount is rejected

TEST SETUP

Try to add NaN oxygen to a facility through the resource API.

EXPECTED RESULT

The facility API rejects the amount with IllegalArgumentException.

18

SIMULATION / INPUT CHECK

Zero simulation time is rejected

TEST SETUP

Ask Simulation to advance a colony by 0 time units.

EXPECTED RESULT

Simulation rejects the input with IllegalArgumentException.

19

MISSION / RESUBMISSION

Repeated mission submission is rejected safely

TEST SETUP

Submit a valid mission once, then submit that same mission object again.

EXPECTED RESULT

The second submission is rejected. The original remains ASSIGNED and active.

OPERATOR WALKTHROUGH / SETUP

Build a small, valid colony

These routes use the visible GUI controls. Start from the example colony and choose distinct names, IDs, and locations.

A Add a crew member

DO THIS

1. Open People & robots. 2. Choose Add crew member. 3. Select a role, enter a name and unused positive Crew ID. 4. Choose health, current base, skills, and fatigue, then save.

EXPECTED

The person appears in the registered list. The crew count increases and the new person is available.

B Add a facility

DO THIS

1. Open Facilities and choose Add facility. 2. Select a facility type and a unique location name. 3. Set capacity, status, and starting supplies. 4. Set power output when creating a Power Station, then save.

EXPECTED

The facility appears in the registry. Its supplies are included in Colony status totals.

C Create and assign a mission

DO THIS

1. Open Missions and choose New mission. 2. Select a mission type and positive duration. 3. Set facility, supplies, required abilities, and type-specific details. 4. Choose the whole team in Step 3; review, then choose Create and assign team.

EXPECTED

A valid mission is marked ASSIGNED and becomes active. Assigned team members show as busy. For Exploration, use a scientist and an equipped exploration rover with the required abilities.

OPERATOR WALKTHROUGH / LIFECYCLE

Run the mission, then test a rejection

The mission controls become available only when the selected active mission is in the matching state.

D

Start, advance, and complete

DO THIS

1. Assign a valid mission with enough required supplies and a duration of at least 3 time units. 2. Choose Start mission. Confirm the status changes to IN PROGRESS. 3. Advance by 1 time unit; verify elapsed time increases but the mission remains active. 4. Choose Complete mission and confirm the action.

EXPECTED

The mission becomes COMPLETED. The active slot clears and its agents become available again. If it completes automatically after reaching its duration, no manual completion is needed.

E

Submit an unsuitable team

DO THIS

1. Choose Missions / New mission / Exploration. 2. Use a positive duration, enough supplies, and an operational facility. 3. In Step 3, select only a mechanical engineer without scientific skill; do not select a scientist or rover. 4. Choose Create and assign team.

EXPECTED

The colony rejects the submission. The GUI explains the missing team requirement and records the rejection in Activity. No mission becomes active; the engineer stays available.

FIELD RECORD / WHAT TO CAPTURE

After a run, note the scenario name, PASS/FAIL, actual result, and any technical details. For hands-on routes, capture the final mission state, active-mission status, and agent availability.

EXPECTED STATES ASSIGNED -> IN PROGRESS -> COMPLETED